

Day(34)

* Flow Control statements in python
(OR)
Control structures in python

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=> purpose of Flow control statements in python.

=> Types of Flow control statements in python.

① Conditional or selection or branching statements

① simple 'if' statement

② if... else statement

③ if... elif... else statement

④ match case statement

② Looping or iterative or repetitive statements.

① while loop or while... else loop

② for loop or for... else loop

③ Transfer flow statements

① break

② continue

③ pass

④ return

=> Combined programming example on conditional, looping & transfer.

=> Inner or nested loops.

① while loop in while loop

② for loop in for loop

③ for loop in while loop.

④ while loop in for loop

==

* Flow control statements in python.

(OR)

Control structures in python

⇒ The purpose of Flow control statement in python is that "to perform an operation / Task one time (X-operation in the case of True or Y-operation in the case of false) OR perform certain operation repeatedly or iteratively for Finite Number of times until Test condition becomes false".

⇒ mobile 300 contacts - invites them Bday --- Looping stmts.
 → one person - - msg of Bday - - - Cnd stmts.
 ⇒ generate 1--100 numbers - - - - Looping stmts.
 ⇒ n is +ve or negative - - - - Cnd. stmts

⇒ In python programming, we have 3 types of Flow control ~~stmts~~ statements. They are:

- ① Conditional OR selection OR Branching statements
- ② Looping OR iterative OR Repetative statements
- ③ Transfer Flow control statements.

① Conditional or selection or Branching statements:-

→ The purpose of Conditional or selection or Branching statements is that "to perform either X-operation in the case of True or perform Y-operation in the case of False only once."

→ In python programming, we have 4 types of Conditional statements are:-

- (I) Simple if statement
- (II) if.... else statement
- (III) if.... elif... else statement
- (IV) match Case statement

(1) simple if statements:-

• Syntax:

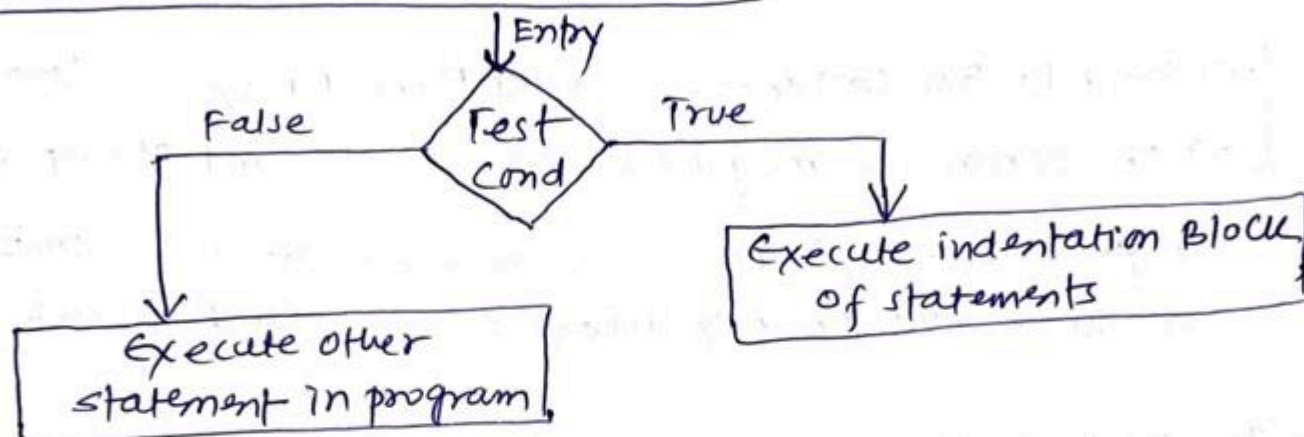
```

if (Test Cond) :
    ...->statement-1
    ...->statement-2
    .....
    ...->statement-n
    other statements in program
  
```

Indentation symbol

Indentation Block of stmt.

• Flow Chart for simple if statements



- Here if is a keyword
- Here Test Cond. can be either True or False
- if the Test Cond is True then pvm executes Execute indentation Block of statements and later pvm executes other statements in program.
- If the Test Cond is False then pvm executes . pvm executes other statements in program without executing indentation Block of statements.

• examples

#movies.

```

tkr=input("Do you have ticket(yes/no):")
if (tkr=="yes"):
    print("enter into theatre")
    print("watch the movies")
    print("eat pop corns")
print("Goto Home").
  
```

tkr.lower()

or use print
or yes, YES.

program for deciding whether the number is even or odd.

```
n=int(input("Enter a Number:"))
```

```
if (n%2 == 0):
    print("{} is EVEN".format(n))
```

```
if (n%2 != 0):
    print("{} is odd".format(n))
```

```
print("program execution completed")
```

$$\begin{array}{r} 2 \overline{) 18} \quad (9 \\ 18 \\ \times \times 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 17} \quad (8 \\ 16 \\ \times 1 \end{array}$$

program for deciding whether the given number is +ve or

```
n=int(input("Enter a Number:"))
```

```
if (n > 0):
    print("{} is +ve".format(n))
```

```
if (n < 0):
    print("{} is -ve".format(n))
```

```
if (n == 0):
    print("{} is zero".format(n))
```

```
print("program execution completed")
```

H/w
palindrome

```
In [3]: #movies
tk=input("do you have a ticket:")
if(tkt=="yes"):
    print("enter into theatre")
    print("watch the movies")
    print("eat pop corn!")
print("goto home")
```

```
enter into theatre
eatch the movies
eat pop corn!
goto home
```

```
In [4]: #ex2.
tk=input("do you have a ticket")
if (tk.lower()=="yes"):
    print('enter into theatre')
    print('watch the movies')
    print('eat pop corn!')
print("Goto home")
```

```
enter into theatre
watch the movies
eat pop corn!
Goto home
```

program for deciding whether the number is odd or even

```
In [6]: n=int(input("enter a number"))
if (n%2==0):
    print("{} is even".format(n))
if (n%2!=0):
    print("{} is odd".format(n))
print("program execution completed")
```

```
17 is odd
program execution completed
```

program for deciding whether the given number is positive or negative

```
In [11]: n=int(input("enter a number:"))
if (n>0):
    print("{} is positive number".format(n))
if (n<0):
    print("{} is neagative number".format(n))
if (n==0):
    print("{} is zero number".format(n))
print("program execution completed")
```

189 is positive number
program execution completed

Write a Python program to check whether a name entered by the user is a palindrome number or not.

```
In [31]: name=input("enter a palindrome")
name_lower=name.lower()
reverse=name_lower[::-1]
if(name_lower==reverse):
    print("{} is a palindrome".format(name))
print("program execution completed")
```

anna is a palindrome
program execution completed

In []: